The child who dreamt of capturing the sun

BY  NINA NOTMAN | 4 MARCH 2026

Organic solar cell windows will enable the buildings of the future to be energy-neutral, says Thuc-Quyen Nguyen

Source: Courtesy Thuc-Quyen Nguyen

Thuc-Quyen Nguyen is an expert in organic electronic devices

Thuc-Quyen Nguyen's interest in harvesting energy from the sun dates back to her childhood in Vietnam. At age five, her mother and siblings moved to a remote village with no electricity or clean water. 'When the sun went down at 5.30–6pm, my mother made us go to bed' as there was no money for oil for a lamp, she says. 'I remember laying there for hours dreaming about capturing [the sunlight] and putting it in a little jar, so that at night I could use it to study or read my book.'

Nguyen was born in 1970 in Buôn Ma Thuột, a city in what was then South Vietnam. Her father was trained by the US to hunt the Viet Cong, and after the Vietnam War ended in 1975, he was imprisoned in a re-education camp. The rest of the family was left destitute and moved to the village to live in a self-built tent. Although life there was difficult, Nguyen credits it with sparking her interest in nature and fostering her creativity, as they had to make everything they needed from foraged materials.

A fresh start

In 1991, her family immigrated to the United States with three sets of clothes and a few words of English through the Humanization Organization programme set up to compensate men imprisoned for fighting alongside the Americans during the Vietnam War. This relocation set Nguyen on the path to becoming an academic – since 2004, she has been a professor of chemistry at the University of California, Santa Barbara. Nguyen says she didn't set out to become a professor, but rather a series of fortuitous decisions led her there. 'In the village, I was raised to be a wife, a daughter and a mother,' she explains.

Related stories

Cryptic chemistry crossword #124

Quick chemistry crossword #102

Chemistry wordoku #133

She initially only enrolled at Santa Monica College for classes to improve her English but later signed up for maths and chemistry classes too. 'The first time I went in the lab, I was so excited,' she says, as she hadn't had access to glassware or chemicals in her village school. After completing a two-year associate degree at the community college, Nguyen transferred to the University of California, Los Angeles (UCLA), to earn a bachelor's degree in chemistry. Her love of chemistry and learning flourished there. 'My nickname at UCLA was "Pourquoi" because I always asked why,' she says.

Not knowing what she wanted to do in the 'real-world', Nguyen applied for – and secured – a place at UCLA to study for a PhD with Benjamin J. Schwartz using ultrafast spectroscopy to study conducting polymers. In 2001, she moved to Columbia University, for a postdoc on molecular electronics with the chemistry Nobel laureate Louis Brus. Next stop was the University of California, Santa Barbara. Organic solar cells were her chosen research focus as a new PI – a nod to her childhood dream of harnessing the sun's rays for electricity, which also used the skillsets she built during her PhD and postdoc.

A window to the future

Source: Courtesy Thuc-Quyen Nguyen

Nguyen passionately supports the next generation of scientists, and helped launch a foundation to reward scientific breakthroughs

Today, Nguyen is a world-leading expert in the development of these and other organic electronic devices. Most solar cells on the market currently are silicon-based, and Nguyen predicts that it will take another decade or more for organic solar cells to become mass produced. Because of their unique properties, they will likely have different uses to silicon-based devices. 'The organic solar cell [is] semi-transparent, lightweight, and you can make it flexible,' Nguyen says. The idea of installing them on windows particularly excites her, as doing so will significantly increase the surface area of buildings that can be covered in solar cells. This is especially true in cities, where there are tall buildings with many windows but little roof space. 'Buildings [currently] consume around 35% of the global energy generation,' she says. Organic solar cells will play a key role in achieving energy neutral buildings, she adds.

Nguyen is committed to supporting the next generation of scientists. Alongside teaching undergraduates and mentoring her research group, she also frequently coaches students from around the world online on how to have an academic career. She has a particular interest in advising immigrants and those experiencing discrimination, something she still experiences today as 'an Asian woman who speaks English with a strong accent'.

In 2020, Nguyen was instrumental in the launch of the VinFuture Foundation, funded by the Vietnamese philanthropists Phạm Nhật Vượng and Phạm Thu Hương. Each year, the foundation awards a $3 million (£2.2 million) VinFuture Prize for scientific breakthroughs with high potential to create meaningful change in people's lives. The 2025 VinFuture Prize, for example, was awarded to the scientists behind vaccines that prevent cancers caused by human papillomavirus. There are also three smaller annual prizes worth $500,000 each. Nguyen is most proud of the awards that recognize innovators from developing countries and women innovators, both of which tend to be overlooked by other international prize communities. She lobbied hard for these categories to be included, she explains.

MORE NINA NOTMAN

The future of total synthesis

AI tools tackle paper mill fraud overwhelming peer review

Rethinking workplaces for neurodivergent staff